Optimization for Deep Learning (OPT4DL)

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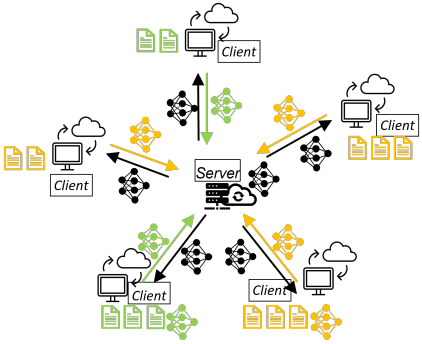
Compression Mechanisms

Distributed Learning

- Agents share model-based representation of local data through (repeatedly) communicating g_t^j with server (or each other)
- For $t = 1, \ldots, T$ Communication/training rounds
  - server sends the current global model $\bar{w}_t$ to the agents
  - each agent computes and sends g_t^j to the server
  - Server aggregates the recieved messages and update the global model

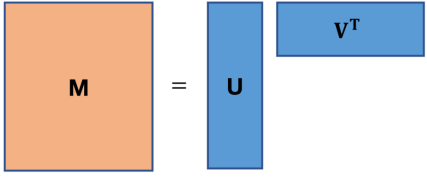
$$\bar{w}_{t+1} = \bar{w}_t - \eta \frac{1}{K} \sum_{j=1}^{K} g_t^j \qquad (1)$$

- Recall $g_t^i \in \mathbb{R}^d$ and we need $\mathcal{O}(32d)$ bits for each message.
- How to do this if agents have low communication resources?



Compression as a Solution

- We will send $\mathcal{C}(g_t^j)$ where $\mathcal{C}(\cdot)$ is a compression mechanism
- Quantization: reduce the precision (bits) of each coordinate from 32 to $q = 1, 2, 4, 8, 16$ bits, saving $\mathcal{O}(32/q)$ in communication resources
- Example: $x = [1.7, -2.6, -3.5]$ and $q = 1$ (sending the signs): $\mathcal{C}(x) = [1, -1, -1]$
- Sparsification: Only communicate k out of the d coordinates (and their indices), saving $\mathcal{O}(d/k)$ in communication resources
- Example: $x = [1.7, -2.6, -3.5]$ and $k = 1$ (selecting the largest entry): $\mathcal{C}(x) = [0, 0, -3.5]$
- Low-rank Factorization: When the gradients are matrices or tensors, find their low rank approximation and communicate that:



Good Compression Mechanisms and Examples



- $\mathcal{C}(g_t^j)$ can be thought of as the estimator of g_t^j. Thus, aa compression is a good one if it has a low estimation error (low bias and variance)

$$\mathbb{E}\|\mathcal{C}(x) - x\|^2 = \underbrace{\mathbb{E}\|\mathcal{C}(x) - \mathbb{E}[\mathcal{C}(x)]\|^2}_{\text{Variance}} + \underbrace{\|\mathbb{E}[\mathcal{C}(x)] - x\|^2}_{\text{Bias}} \tag{2}$$

  where $\mathbb{E}$ is w.r.t. the random of $\mathcal{C}$

- Random Sparsification: Uniformly at random select k out of d coordinates.
  - It turns out $\mathbb{E}[\mathcal{C}(x)] = \frac{k}{d}x$ so we have some bias
  - It turns out $\mathbb{E}\|\mathcal{C}(x) - x\|^2 \leq (1 - \frac{k}{d})\|x\|^2$
  - Can be made unbiased by using $\tilde{\mathcal{C}}(x) = \frac{d}{k}\mathcal{C}(x)$
  - But, doing so results in variance blowup: $\mathbb{E}\|\mathcal{C}(x) - x\|^2 \leq (\frac{d}{k})^2\|x\|^2$
  - Takeaway 1: The biased version seems to be better in terms of estimation error
  - Takeaway 2: For both versions, the error vanishes as $x \to 0$, happens when x is (an estimator of) the gradient.

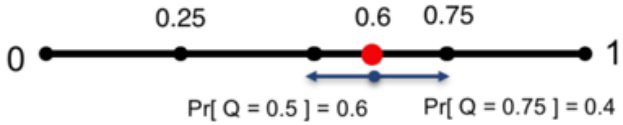
Good Compression Mechanisms and Examples (Cont'd)



- Top-k Sparsification: select k largest out of d coordinates
  - A deterministic operator and thus will have a bias
  - should be no worse that random sparsification so $\|\mathcal{C}(x) - x\|^2 \leq (1 - \frac{k}{d})\|x\|^2$
  - the error vanishes as $x \to 0$, happens when x is (an estimator of) the gradient.
- Random Dropout: set $\mathcal{C}(x) = x$ with probability p (full communication) and set $\mathcal{C}(x) = 0$ with probability $1 - p$ (no communication)
  - It turns out $\mathbb{E}[\mathcal{C}(x)] = px$ so we have some bias
  - It turns out $\mathbb{E}\|\mathcal{C}(x) - x\|^2 \leq (1 - p)\|x\|^2$
  - Can be made unbiased by using $\tilde{\mathcal{C}}(x) = \frac{1}{p}\mathcal{C}(x)$
  - But, doing so results in variance blowup: $\mathbb{E}\|\mathcal{C}(x) - x\|^2 \leq (\frac{1}{p})^2\|x\|^2$
  - Takeaway 1: The biased version seems to be better in terms of estimation error
  - Takeaway 2: For both versions, the error vanishes as $x \to 0$, happens when x is (an estimator of) the gradient.
  - Takeaway 3: on average, random dropout and random sparsification are identical with $p = k/d$

Random Quantization

- Normalize x by forming $u = \frac{x}{\|x\|} \odot \text{sign}(x)$
- Each coordinate of u is in $[0, 1]$ and we can quantize each separately and randomly into q levels to obtain $\tilde{u}$



- Send $\mathcal{C}(x)\|x\| \odot \text{sign}(x) \odot \tilde{u}$
  - It can be shown it is unbiased $\mathbb{E}[\mathcal{C}(x)] = x$
  - The variance satisfies $\mathbb{E}\|\mathcal{C}(x) - x\|^2 \leq \min\{\frac{d}{s^2}, \frac{\sqrt{d}}{s}\}\|x\|^2$
  - q-bit random quantization saving us $\mathcal{O}(32/q)$ bits in communication
  - the error vanishes as $x \to 0$, happens when x is (an estimator of) the gradient.
  - We can rescaled it to reduce the variance at the cost of some bias.

Compression and Convergence



- Consider a compression mechanism such that

$$\mathbb{E}[\mathcal{C}(x)] = Ax, \qquad \mathbb{E}\|\mathcal{C}(x) - x\|^2 \leq B\|x\|^2, \qquad 0 < A \leq 1, \;\; B \geq 0 \tag{3}$$

- For instance, $A = p$ and $B = 1 - p$ for random dropout. Or $A = 1$ and $B = \frac{d^2}{k^2}$ for unbiased random sparsification.
- Consider the following key step for SGD with compression, i.e. $w_{t+1} = w_t - \eta \mathcal{C}(g_t)$

$$\begin{aligned} f(w_{t+1}) &\leq f(w_t) + \langle w_{t+1} - w_t, \nabla f(w_t)\rangle + \frac{L}{2}\|w_{t+1} - w_t\|^2 \\ &= f(w_t) - \eta\langle \mathcal{C}(g_t), \nabla f(w_t)\rangle + \frac{L\eta^2}{2}\|\mathcal{C}(g_t)\|^2 \end{aligned} \tag{4}$$

- Taking expectation w.r.t. to $\mathcal{C}$ and then the randomness of current batch z_t conditioning on previous batches

$$\mathbb{E}[f(w_{t+1})] \leq f(w_t) - \eta A\|\nabla f(w_t)\|^2 + \frac{L\eta^2}{2}\mathbb{E}\|\mathcal{C}(g_t)\|^2 \tag{5}$$

Compression and Convergence (Cont'd)



- Note that

$$\mathbb{E}\|\mathcal{C}(g_t)\|^2 = \mathbb{E}\|\mathcal{C}(g_t) \pm g_t\|^2 = \mathbb{E}\|\mathcal{C}(g_t) - g_t\|^2 + \mathbb{E}\|g_t\|^2 + \mathbb{E}[\langle \mathcal{C}(g_t) - g_t, g_t\rangle] \tag{6}$$

- First term on RHS is bounded by $\mathbb{E}\|\mathcal{C}(g_t) - g_t\|^2 \leq B\mathbb{E}\|g_t\|^2$
- Third term on RHS can be handled as follows

$$\mathbb{E}[\langle \mathcal{C}(g_t) - g_t, g_t\rangle] = \mathbb{E}[\langle \mathcal{C}(g_t) - g_t \pm Ag_t, g_t\rangle] = \mathbb{E}[\langle (1-A)g_t, g_t\rangle] = (1-A)\mathbb{E}\|g_t\|^2 \tag{7}$$

  assuming $\mathcal{C}$ is independent of z_t and using $\mathbb{E}[\mathcal{C}(g_t)] = Ag_t$.
- Also recall using the SFO assumption $\mathbb{E}\|g_t\|^2 \leq \sigma^2 + \|\nabla f(w_t)\|^2$
- All in yields

$$\mathbb{E}[f(w_{t+1})] \leq f(w_t) - \eta\Big(A - \frac{L\eta}{2}(B+2-A)\Big)\|\nabla f(w_t)\|^2 + \frac{L\eta^2}{2}(B+2-A)\sigma^2 \tag{8}$$

Compression and Convergence (Cont'd)



- The rest is similar to SGD's analysis and we can obtain

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}_{\mathbf{z_{1:T}}}\|\nabla f(w_t)\|^2 \leq \frac{f(w_1)-f^*}{T\eta\left(A-\frac{L\eta}{2}P\right)}+\frac{L\eta\sigma^2 P}{2A-L\eta P}, \qquad P:=(B+2-A) \tag{9}$$

- We now require a smaller learning rate for the result $\eta < \frac{2A}{LP}$ (vs. $\eta < 2/L$)
- There is effectively a new variance: $\tilde{\sigma}^2 = \sigma^2(B+2-A) \geq \sigma^2$. For instance:
  - for random dropout where $A = p$ and $B = 1-p$, $\tilde{\sigma}^2 = (3-2\rho)\sigma^2$
  - For unbiased random sparsification where $A = 1$ and $B = \frac{d^2}{k^2}$, $\tilde{\sigma}^2 = \sigma^2(1+\frac{d^2}{k^2})$
- The result reduces to SGD with no compression when $A = 1$ and $B = 0$.
- We can show $\mathbb{E}\|\nabla f(\hat{w})\|^2 \simeq \mathcal{O}(A^{-1}/\sqrt{T})$ when $A < 1$ and $B < 1$ (biased operators) while $\mathbb{E}\|\nabla f(\hat{w})\|^2 \simeq \mathcal{O}(B/\sqrt{T})$ when $A = 1$ and $B > 1$ (unbiased operators).
- Still converges, but now the dominant term is much worse as A^{-1} and B are very large (e.g. $\mathcal{O}(\frac{d}{k})$ and $\mathcal{O}(\frac{d^2}{k^2})$). Also, generally bias seems helpful

Error Feedback



- Let us focus on a biased compression operator such that $\mathbb{E}\|\mathcal{C}(x) - x\|^2 \leq (1-\delta)\mathbb{E}\|x\|^2$. This is called a contraction compression.
  - $\delta \to 1$ low compression (or a good operator)
  - $\delta \to 0$ high compression (or a bad operator)
- One way to interpret compression is that we are throwing out part of the information. How about we save it in memory, augment it with the next message, and the communicate that?
- In some sense, instead of throwing information out, we are communicating them with delay.
- And if we communicate long enough, everything more or less will be sent.
- Mathematically,

$$\text{communicate} \quad \mathcal{C}(x_t + e_t) \quad \text{instead of} \quad \mathcal{C}(x_t), \qquad e_{t+1} = (x_t + e_t) - \mathcal{C}(x_t + e_t) \tag{10}$$

- e_{t+1} is the compression error, $x_t + e_t$ is the uncompressed message, and $\mathcal{C}(x_t + e_t)$ is what we send out.

The EF-SGD Algorithm



- Initialize w_1 and $e_1 = 0$
- For $t = 1, \ldots, T$ do
  - Sample a batch $z_t \sim p_z$
  - Form the stochastic gradient $g_t = \nabla \ell(w_t, z_t)$
  - Compress and send $m_t = \mathcal{C}(g_t + e_t)$
  - Update the error $e_{t+1} = (g_t + e_t) - \mathcal{C}(g_t + e_t)$ and store it
  - Update the parameters $w_{t+1} = w_t - \eta m_t$

Theorem

Assume f is L-smooth and we have access to an SFO and that $z_1, \ldots, z_T$ are statistically independent. Then, If $\eta = \mathcal{O}(1/\sqrt{T})$, EF-SGD achieves $\mathbb{E}\|\nabla f(\hat{w})\|^2 \leq \mathcal{O}(\frac{1}{\delta T} + \frac{1}{\sqrt{T}})$.

- better dependence on the compression parameter as it appears only in the higher order term (as opposed to $\mathcal{O}(\frac{1}{\delta\sqrt{T}})$ for SGD without EF).

Proof Sketch



- Recall for decentralized SGD we considered how the average model (a virtual sequence unknown to the agents) performs. Here we also consider an idealized sequence with no compression error
- Define $\tilde{w}_t = w_t - \eta e_t$.
- Effectively removing the compression error from EF-SGD's iterates $w_{t+1} = w_t - \eta \mathcal{C}(g_t + e_t)$
- Intuitively, $\tilde{w}_t$ should resemble SGD's update with no compression, i.e.

$$\tilde{w}_{t+1} = \tilde{w}_t - \eta g_t \tag{11}$$

- Indeed that is the case:

$$\tilde{w}_{t+1} = w_{t+1} - \eta e_{t+1} = w_t - \eta \mathcal{C}(g_t + e_t) - \eta e_{t+1} = (w_t - \eta e_t) - \eta g_t = \tilde{w}_t - \eta g_t \tag{12}$$

where we used EF-SGD's update and $e_{t+1} = (g_t + e_t) - \mathcal{C}(g_t + e_t)$

Proof Sketch (Cont'd)



- We then use our potential-based analysis as before.
- We consider and analyze the evolution of the compression error $e_{t+1} = (g_t + e_t) - \mathcal{C}(g_t + e_t)$ and obtain

$$\mathbb{E}\|e_t\|^2 \leq \kappa(\delta)\mathbb{E}\|e_{t-1}\|^2 + \text{ other terms} \tag{13}$$

  for some $0 < \kappa(\delta) < 1$ depending only on the compression parameter
- We then define a potential function

$$\Phi_t = \mathbb{E}[f(\tilde{w}_t)] - f^* + C_e\mathbb{E}\|e_t\|^2 \tag{14}$$

- Study the evolution of potential by summing over its consecutive differences
- When $\kappa(\delta) < 1$ we can find a good constant C_e to ensure convergence.